OOPs and Exception Handling

1. What is Object Oriented Programming? Explain class and object with a simple Python example.

```
class myclass():
  x = 33

a = myclass()
b = myclass()
print(a.x)
print(b.x)
```

```
33
33
```

2. Explain the role of the **init** method in a class. When is it called?

ans - **init** is called automatically when class is initiated

```
class myclass():
  def __init__(self, a,b):
    self.a = a
    self.b = b
p = myclass(10, 20)
print(p.a + p.b)
```

```
30
```

3. What is encapsulation? How does Python achieve data hiding? Explain with an example.

```
class Person:
  def __init__(self, name, age):
    self.name = name
    self.__age = age

  def get_age(self):
    return self.__age

  def set_age(self, age):
    if age > 0:
      self.__age = age
    else:
      print("Age must be positive")

p1 = Person("Tobias", 25)
print(p1.get_age())
```

```
p1.set_age(26)
print(p1.get_age())
```

```
25
26
```

4. What is inheritance? Write a Python program to show single inheritance.

```
class Person:
  def __init__(self, fname, lname):
    self.firstname = fname
    self.lastname = lname

  def printname(self):
    print(self.firstname, self.lastname)

class Student(Person):
  pass
x = Student("Mike", "Olsen")
x.printname()
```

```
Mike Olsen
```

5. What is an exception in Python? Why is exception handling required?

```
def div(a,b):
  try:
    res = a // b
    print(res)
  except ZeroDivisionError:
    print("Error")
div(10,5)
div(10,0)
```

```
2
Error
```

6. Write a Python program to handle division by zero and invalid input using exception handling.

```
def div ():
  try:
    a = float(input("Enter Numerator : "))
    b = float(input("Enter Denominator : "))

    res = a // b
  except ZeroDivisionError:
    print(f"Error! You cannot divide by zero.\n")
  except ValueError:
    print("Error: Invalid input. Please enter numeric values only.\n")
  except Exception as e:
```

```
        print(f"An unexpected error occurred: {e} \n")
    else:
        print(f"Success! The result is: {res}\n")

div()
div()
div()
```

```
Enter Numerator : 10
Enter Denominator : 5
Success! The result is: 2.0

Enter Numerator : 10
Enter Denominator : 0
Error! You cannot divide by zero.

Enter Numerator : aa
Error: Invalid input. Please enter numeric values only.
```

Start coding or generate with AI.